Stochastic Discrete Local Volatility

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Abstract

We present a fully arbitrage-free option market model using the SANOS [6] framework based in ideas of [19].

The model is the product of a risk-neutral discrete state martingale defined on a fixed grid with stochastic "discrete local volatilities", and a simple SANOS-type Black & Scholes smoothing process which allows smooth interpolation of option prices between grid points.

We train the model on S&P 500 Index options data.

Abstract

We construct an option market model on a fixed lattice of cash strikes and expiries that is free of static and dynamic arbitrage by construction. The state is the vector of gridded risk-neutral densities, encoded by discrete local variances: their positivity is equivalent to the absence of static arbitrage, so in log coordinates the state space is Euclidean. Absence of dynamic arbitrage then fixes the drift node by node, as a covariance between the volatility of each discrete local variance and the relative volatility of the density at the same node. The restriction goes back to Carmona and Nadtochiy [9] in the continuum and to Wissel [19] on this lattice; on a fixed lattice it becomes an algebraic identity, free of the regularity conditions and the maturity derivative of the continuum, and is computed by one tridiagonal solve per factor and expiry. The spot is read off the densities, and at each expiry the front density can be made to collapse onto a node, so every gridded contract settles at its exact payoff. Our main result is existence: for any bounded, locally Lipschitz volatility structure and any number of factors, the model has a unique strong solution up to the first expiry and, restarting at each expiry, across all expiries, with no conditions on the wings. The key estimate bounds the growth of the relative density volatility $\eta = \psi/q$ across expiries by a constant of the grid alone, turning an assumption of Wissel into a theorem.

1 Introduction

This article presents an arbitrage-free market model for for smooth surfaces of European option prices.

The core architecture follows the SANOS approach [6] which models the stock price as the product of a time-and-state discrete process X and a smoothing process Y, in our case Black & Scholes. The former is used to encode the option market using a discrete form of local volatilities as code book. That does not mean that the model is a local volatility model – we merely use the (discrete) local volatilities as a convenient representation of the option surface. Compared to using implied volatilities the advantage is that mere positivity of the local volatilities is sufficient to ensure the the option market is free of static arbitrage at all times. We have shown

in [6] that this representation can be fitted to real life market data very efficiently using linear programming.

In [7] we have shown how to estimate the dynamics such (discrete) local volatilities of such a model under the statistical measure $\mathbb{P}$ which fit very well to real-life observed option bid/asks.

The key contribution in this paper is showing how to construct a corresponding risk-neutral measure $\mathbb{Q}$ or, equivalently, the risk-neutral drift of the model. Our approach is closely related to the seminal but unpublished work [19].

An option market model reverses the usual direction of construction: instead of specifying a process for the underlying and deriving option prices from it, one models the code-book of prices directly, so that a calibration is a change of coordinates rather than an inverse problem. Two conditions must then be maintained at once. The state must remain free of *static* arbitrage, and its dynamics must admit one equivalent local martingale measure common to the underlying and to every traded option. The second is the option analogue of an HJM drift restriction: the volatility structure may be specified freely, and the drift is then determined. The lattice here is *finite*, finitely many strikes and finitely many expiry dates, which is what is traded.

What is already known. Schweizer and Wissel treat the problem in the continuum in two companion papers, one in each direction of the surface: [16] varies maturity at a single strike, and [15] varies strike at a single maturity. In each the code-book is chosen so that its positivity encodes absence of arbitrage, the drift restriction is derived, and existence is proved. The multi-strike multi-maturity case, which is ours, is stated as open in [15], with the expectation that a new parametrisation will be needed. Carmona and Nadtochiy [9] give a continuum local-volatility code-book with a drift restriction, but no existence theorem. On a finite lattice, and answering the multi-maturity question, stands Wissel [19]; his construction is the one this paper is closest to, and it is described separately below.

Modern approaches. The question was reopened from the data side, by the scarcity of option-market histories for training and evaluating hedging strategies [5]. Cohen, Reisinger and Wang [10] build neural stochastic differential equations on a floating lattice and penalise a least-squares residual in place of the drift restriction. [17] simulate an equity option book with a generative adversarial network whose state is the discrete local variance grid compressed by principal components, so that static arbitrage is absent by positivity of that grid; that is the code-book choice this paper inherits, and [18] extend it to a multi-asset universe. [7] decode statically arbitrage-free surfaces from a low-dimensional code, fit a linear state process to it, and test for dynamic arbitrage after the fact, reporting it in 12.5% of states at zero transaction cost.

None of the four enforces the drift restriction exactly, and that is where this paper starts: the cross-sectional constraints govern each date's surface and its terminal payoff cone, and leave the transition of a fixed contract between two dates unconstrained. The choice of grid is part of the same point. All four index their nodes relative to the spot rather than in fixed cash terms, and on such a grid a node is not one contract through time, so there is no single instrument whose price is required to be a martingale and nothing for an HJM-type restriction to constrain. Fixing the strikes in cash is what buys the restriction here.

The closest prior work. Wissel [19] works on a finite lattice of fixed cash strikes and fixed expiry dates, parametrises it by a price level together with positive local implied volatilities, derives the no-arbitrage drift restrictions, and proves constructive strong existence and uniqueness

for a class of coefficients. Behind the front expiry his local implied variance is *exactly* the backward discrete local variance used here, an algebraic identity on the grid, and his drift restriction is, after a change of measure, exactly the one we state in Theorem 3.2. We claim no priority for it. That working paper appears never to have been published in archival form.

What this paper does

We keep the same lattice and change the chart: the state is the vector of gridded *densities* together with the discrete local variances, rather than prices together with quantiles of digitals. Three consequences follow, and they are the contributions.

1. **The spot is a read-off, and it settles exactly.** The spot is $S_t = K^\top q_t^j$ at every expiry j (Proposition 3.3), an identity of the chart holding at all t. At the front expiry Proposition 4.2 exhibits a family of front volatilities under which q^1 collapses onto a single node, so the spot lands on a strike and every gridded contract settles at its exact payoff (Section 4.2). That pinning is a property of the volatility chosen and not of the chart: by Remark 4.3 the terminal states the chart admits carry every spot in the grid range, whereas in [19] the pinning follows from the chart whatever the coefficients. We do not construct dynamics reaching an off-lattice terminal spot; that is left as future work.

2. **Existence, with no condition on the wings.** For every bounded and locally Lipschitz volatility structure b and any number of driving factors, the model has a unique strong solution on $[0, T]$ for every $T < T_1$, positive by construction; it extends to T_1 (Theorem 5.6(ii)) and, restarting at each expiry with the settling front, to the whole ladder $[0, T_M]$, unique given that front (Corollary 5.8). The quantity that must be controlled is $\eta = \psi/q$, the density volatility relative to the density, and its growth across expiries is bounded by the grid alone: Theorem 5.4 gives $\Theta_j \le N + 1$ uniformly in the state and in the volatility structure. *Wissel [19] must assume a bound of this kind; here it is a theorem.* The front expiry costs one hypothesis, and it is a mild one: η^1 must be integrable in t, not bounded, because η^j is affine in η^1 and an Itô equation needs an integrable drift rather than a bounded one. The front volatility used here diverges at rate $(T_1 - t)^{-p}$, and the two demands made of p leave a window: $p < 1$ is what Assumption 5.5 asks, while $p \ge \frac{1}{2}$ is what makes the front density collapse, so that every gridded contract settles at its exact payoff (Proposition 4.2).

3. **The drift is pointwise, and the count works out.** In density coordinates the drift equation is $N + 2$ conditions on N unknowns, and is consistent because the left null space of $\widehat{G}_j$ is exactly $\operatorname{span}\{\mathbf{1}, K\}$, conservation of mass and of the spot (Lemma 2.3). The state-independent factor then cancels from both sides and the drift is available nodewise, $\alpha^{j,i} = -\langle \eta^{j,i}, \beta^{j,i} \rangle$, with no matrix inversion and no rank condition.

Scope, and what is not claimed

The over-determination of the third contribution is not intrinsic: it is created by the choice of density coordinates and removed by the same choice, and in price coordinates the system is square and the question does not arise.

The existence statement is for a fixed finite grid. The constant is $(M-1)(N+1)$ and depends on the grid. The one hypothesis, Assumption 5.5, is a condition on the front volatility and not on the state, and it is sufficient rather than necessary.

2 The code-book

We assume deterministic forwards and discount factors, which reduces to a driftless spot with unit discount factors, cf. [4]. The spot at $t = 0$ is S_0.

We follow the [6] architecture for a time-homogeneous fixed strike grid: Fix model expiries $0 =: T_0 < T_1 < \cdots < T_M$ with $\Delta_j := T_j - T_{j-1}$, and a single strike grid $0 \le K^0 < K^1 < \cdots < S_0 < \cdots < K^N < K^{N+1}$, held fixed in cash terms and common to all expiries, with largest spacing $h := \max_i(K^{i+1} - K^i)$; the grid need not be uniform. Let $q = (q^{j,i})$ with $q^j \in \mathbb{R}^{N+2}$ be a discrete risk-neutral density in the sense that

1. $q^{j,i} \ge 0$
2. $\sum_{i=0}^{N+1} q^{j,i} = 1$
3. $\sum_{i=0}^{N+1} K^i q^{j,i} = S_0$.

We denote by $\Sigma_t^{j,i}$ the discrete local variance (the squared discrete local volatility) at the interior node $i = 1, \dots, N$ and expiry j, as seen at calendar time t, and by $\Sigma_t^j \in (0,\infty)^N$ the vector of them, with $\lambda = \log \Sigma$ throughout; it enters the code-book multiplied by the slab length Δ_j. Prices are the linear functional $C_t(K^i, T_j) = (\pi^i)^\top q_t^j$ with π^i the discretised payoff $(s - K^i)^+$.

If it's the same for everyone I'd align the notation with DYSANOS, e.g. use $\hat{T}$ and $\hat{K}$ for relative model coordinates, and T, K for the market. I'd generally use the same terms. I would also write everything in discrete local vol-squared and specifically add the "dt" term because otherwise the numbers have quite different dimensionality.

Propagating densities rather than prices, in Fokker–Planck form, is what makes the systems below tridiagonal. With the half-weights

$$\omega^k := \frac{(K^k)^2}{K^{k+1} - K^{k-1}}, \qquad \iota^{k,-} := \frac{\omega^k}{K^k - K^{k-1}}, \qquad \iota^{k,+} := \frac{\omega^k}{K^{k+1} - K^k}, \qquad k = 1, \dots, N, \tag{1}$$

the recursion is $A_j(\Sigma^j)\, q^j = q^{j-1}$, where $A_j(\Sigma)$ is tridiagonal with rows (here q^i denotes the i-th component of a generic $q \in \mathbb{R}^{N+2}$)

$$\big(A_j(\Sigma)q\big)_i = \begin{cases} q^0 - \Delta_j \Sigma^1 \iota^{1,-} q^1, & i = 0, \\ -\Delta_j \Sigma^{i-1} \iota^{i-1,+} q^{i-1} + \big(1 + \Delta_j \Sigma^i (\iota^{i,-} + \iota^{i,+})\big) q^i - \Delta_j \Sigma^{i+1} \iota^{i+1,-} q^{i+1}, & 1 \le i \le N, \\ -\Delta_j \Sigma^N \iota^{N,+} q^N + q^{N+1}, & i = N+1. \end{cases} \tag{2}$$

Up to notation this is the implicit forward operator of [8]. Note that in (2) the entry multiplying $q^{i\pm1}$ carries $\Sigma^{i\pm1}$ and not Σ^i.

Because (2) is affine in Σ we may write

$$A_j(\Sigma)\, q \;=\; q + G_j(q)\, \Sigma, \qquad G_j(q) \in \mathbb{R}^{(N+2)\times N}, \tag{3}$$

so that $G_j(q)$ collects the terms carrying Σ. The map G_j is linear in q and $G_j(q)\Sigma$ is linear in each argument separately; we use both readings. The following elementary observation is used constantly.

Lemma 2.1 (Factorisation). *Column k of $G_j(q)$ involves q only through q^k. Consequently*

$$G_j(q) \;=\; \widehat{G}_j \operatorname{diag}(q^1, \dots, q^N), \tag{4}$$

where $\widehat{G}_j \in \mathbb{R}^{(N+2)\times N}$ *has k-th column* $\Delta_j\big(-\iota^{k,-}e_{k-1} + (\iota^{k,-} + \iota^{k,+})e_k - \iota^{k,+}e_{k+1}\big)$. *Moreover* $\widehat{G}_j$ *has rank* N, *hence is injective.*

Proof. Immediate from (2), whose terms carrying Σ^k are all proportional to q^k. For injectivity, suppose $\widehat{G}_j c = 0$ and let k be the smallest index with $c_k \neq 0$. Row $k-1$ receives a contribution from column k' only for $k' \in \{k-2, k-1, k\}$, and the first two carry $c_{k'} = 0$. So row $k-1$ reads $-\Delta_j \iota^{k,-} c_k = 0$, forcing $c_k = 0$, a contradiction. □

Note that $\widehat{G}_j$ depends only on the grid and on Δ_j, not on the state; the state enters (4) only through the diagonal factor.

Proposition 2.2 (The code-book)**.** *Fix the front density* q^1, *a probability vector on the grid with mean* S_0, *and let* $j = 2, \dots, M$.

(i) Given densities q^{j-1} *and* q^j *with* $q^j > 0$, *the discrete local variance is uniquely determined by the linear system*

$$G_j(q^j)\,\Sigma^j \;=\; q^{j-1} - q^j. \tag{5}$$

(ii) Conversely, given $\Sigma^j > 0$, *the vector* $q^j := A_j(\Sigma^j)^{-1} q^{j-1}$ *is again a probability vector with mean* S_0, *and the gridded prices* $(\pi^i)^\top q^j$ *admit no static arbitrage.*

(iii) A gridded surface is free of static arbitrage, butterflies and calendar spreads together, if and only if it arises as in (ii) *from some* $\Sigma \geq 0$, *and it lies in the* interior *of the arbitrage-free set if and only if* $\Sigma > 0$.

Proof. (i) is (3), with uniqueness from Lemma 2.1. For (ii), $A_j(\Sigma^j)^{-1}$ is non-negative with unit column sums and inherits the martingale property [8], attributed there to Andreasen and Huge [1]; non-negativity excludes butterfly arbitrage, and q^j is then a mean-preserving spread of q^{j-1}, so Jensen's inequality excludes calendar arbitrage. (iii) is [8], after [3]: absence of arbitrage is equivalent to non-negative Gamma and Backward-Theta at every node, and Σ is their ratio. Both conditions are nodewise, and $\Sigma^{j,i} = 0$ exactly when Backward-Theta vanishes. □

What everything else rests on is that the admissible set is a product of half-lines, one per node, so it has no faces in the sense that a polytope does, and $\lambda = \log \Sigma$ maps its interior diffeomorphically onto $\mathbb{R}^{N(M-1)}$.

2.1 Two conservation laws

The whole structure of the drift restriction follows from one observation about the left null space of G_j. Both identities below are used by [8], where they hold by construction for the backward operator. What is not there, because nothing in that paper needs it, is that $\operatorname{span}\{\mathbf{1}, K\}$ is the *entire* left null space. That is a rank statement, and it is what makes the drift system of Section 3 square.

Lemma 2.3 (Left null space)**.** *For every* q *and every* j,

$$\mathbf{1}^\top G_j(q) = 0 \qquad \textit{and} \qquad K^\top G_j(q) = 0,$$

where $\mathbf{1} = (1, \dots, 1)$ *and* $K = (K^0, \dots, K^{N+1})$. *Consequently* $A_j(\Sigma)$ *conserves both total mass and the spot, for any* Σ. *Combined with Lemma 2.1, the left null space of* $\widehat{G}_j$ *is exactly* $\operatorname{span}\{\mathbf{1}, K\}$.

Proof. Fix an interior index k and collect the three entries of $G_j(q)$ in column k: from (2) they are $\Delta_j(\iota^{k,-}+\iota^{k,+})q^k$ in row k, $-\Delta_j\iota^{k,+}q^k$ in row $k+1$, and $-\Delta_j\iota^{k,-}q^k$ in row $k-1$. Their sum is 0, giving the first identity. Weighting by the node values,

$$\begin{aligned}\Delta_j q^k\Big[K^k\big(\iota^{k,-}+\iota^{k,+}\big)-K^{k+1}\iota^{k,+}-K^{k-1}\iota^{k,-}\Big] &= \Delta_j q^k\Big[\iota^{k,-}(K^k-K^{k-1})-\iota^{k,+}(K^{k+1}-K^k)\Big]\\ &= \Delta_j q^k\big[\omega^k-\omega^k\big]=0\end{aligned}$$

by (1), giving the second. The conservation statements follow from (3): $\mathbf{1}^\top A_j(\Sigma)q = \mathbf{1}^\top q$ and $K^\top A_j(\Sigma)q = K^\top q$. Since $\widehat{G}_j$ has rank N and lives in $\mathbb{R}^{(N+2)\times N}$, its left null space is two-dimensional, hence exactly $\operatorname{span}\{\mathbf{1}, K\}$. □

Remark 2.4 (Counting). For $q>0$, $G_j(q)$ has $N+2$ rows and N columns and rank N. Hence the $N+2$ martingale conditions per expiry contain exactly two redundancies, conservation of mass and of the spot, and the remaining N conditions match the N unknowns of Σ^j. The system is square on the quotient.

3 The drift restriction

Assumption 3.1 (Martingale property)**.** Under $\mathbb{Q}$ every gridded instrument price is a martingale in t. Since the gridded payoffs π^i together with the unit payoff span $\mathbb{R}^{N+2}$, and $\mathbf{1}^\top q^j = 1$, this is equivalent to each density vector being a martingale,

$$\mathrm{d}q_t^j = \psi_t^j\,\mathrm{d}W_t, \qquad \psi_t^j \in \mathbb{R}^{(N+2)\times d}. \tag{6}$$

This is the only assumption behind the drift restriction, and it carries no regularity. Each q^j is a fixed linear combination of traded butterflies, so (6) is the martingale representation theorem applied to traded prices, and nothing has to commute with a stochastic integral. This is the substantive advantage of a discrete code-book over its continuum analogue, where the corresponding step requires differentiating a stochastic field in strike and maturity and must be licensed by standing regularity conditions. Theorem 3.2 below needs only $q^j>0$.

Theorem 3.2 (Drift restriction)**.** *Let* $\mathrm{d}\Sigma_t^j = \alpha_t^j\mathrm{d}t + \beta_t^j\mathrm{d}W_t$ *and* $q^j>0$. *Under Assumption 3.1, for each* $j=2,\dots,M$,

$$A_j(\Sigma^j)\,\psi^j \;=\; \psi^{j-1} - G_j(q^j)\,\beta^j, \tag{7}$$

marched forward in j *from* ψ^1*, the volatility of the front density, which is given by its chart (Section 4). With the relative density volatility*

$$\eta^{j,i} \;:=\; \frac{\psi^{j,i}}{q^{j,i}}, \qquad i=0,\dots,N+1, \tag{8}$$

the drift is determined node by node,

$$\alpha^{j,i} \;=\; -\langle\eta^{j,i},\beta^{j,i}\rangle \qquad\Longleftrightarrow\qquad \mathrm{d}\lambda^{j,i} = \Big[-\langle\eta^{j,i},b^{j,i}\rangle - \tfrac{1}{2}|b^{j,i}|^2\Big]\mathrm{d}t + b^{j,i}\,\mathrm{d}W_t, \tag{9}$$

where $\lambda=\log\Sigma$ *and* $b=\beta/\Sigma$.

Proof. By Lemma 2.1, $G_j(q)\Sigma = \widehat{G}_j(\Sigma \circ q)$, where $\Sigma \circ q := (\Sigma^i q^i)_{i=1}^N$ is the entrywise product on the interior nodes. The code-book therefore reads

$$q_t^j + \widehat{G}_j\, p_t^j = q_t^{j-1}, \qquad p^{j,i} := \Sigma^{j,i} q^{j,i}, \quad i = 1, \dots, N,$$

with $\widehat{G}_j$ constant. Itô's product rule for the scalar processes $\Sigma^{j,i}$ and $q^{j,i}$ gives

$$\mathrm{d}p^{j,i} = \Big[q^{j,i}\alpha^{j,i} + \langle \psi^{j,i}, \beta^{j,i}\rangle\Big]\mathrm{d}t + \big(\Sigma^{j,i}\psi^{j,i} + q^{j,i}\beta^{j,i}\big)\mathrm{d}W_t, \tag{10}$$

with $\psi^{j,i}, \beta^{j,i} \in \mathbb{R}^{1\times d}$ the rows at node i. Hence $\mathrm{d}q^j + \widehat{G}_j\,\mathrm{d}p^j = \mathrm{d}q^{j-1}$, and by uniqueness of the decomposition of a continuous semimartingale the $\mathrm{d}W$ and $\mathrm{d}t$ terms match separately. The $\mathrm{d}W$ terms give $\psi^j + \widehat{G}_j\big(\Sigma^j \circ \psi^j + q^j \circ \beta^j\big) = \psi^{j-1}$, factor by factor, which is (7) by (3). Since q^j and q^{j-1} are driftless by Assumption 3.1, the $\mathrm{d}t$ terms give

$$\widehat{G}_j\Big(q^j \circ \alpha^j + \textstyle\sum_f \psi^{j,\cdot,f} \circ \beta^{j,\cdot,f}\Big) = 0.$$

Since $\widehat{G}_j$ is injective (Lemma 2.1) the bracket vanishes, $q^{j,i}\alpha^{j,i} + \langle \psi^{j,i}, \beta^{j,i}\rangle = 0$ node by node, which is (9) after dividing by $q^{j,i} > 0$. The logarithmic form is Itô applied to $\lambda = \log\Sigma$, using $\alpha/\Sigma = -\langle \eta, b\rangle$. □

Only the interior components of η enter (9), since Σ lives on the interior nodes; the boundary components are needed in Section 5.

Equation (7) is not ours. Compared with the pricing recursion $A_j(\Sigma^j)q^j = q^{j-1}$ it is *the same operator with one added source*, the structure of Carmona and Nadtochiy [9]. The drift restriction above is theirs in the continuum, and Wissel's [19] on this lattice; we claim no priority for either.

What the fixed lattice changes is threefold. Their restriction carries a term in the derivative of the code-book along time to maturity; a lattice of fixed expiry dates has no such derivative, and here that term is absent. The relation is algebraic rather than differential, so neither the high-order regularity their statement requires nor the singular behaviour as time to maturity vanishes that it is built to accommodate can arise here, the front slab having positive width. And the discrete system is over-determined, $N + 2$ conditions on N unknowns, which is what the counting of Remark 2.4 squares.

Two features of (9) are worth drawing out. The first is its shape. It is the exact analogue of the HJM restriction $\alpha = \sigma \int \sigma$: the drift of the log discrete local variance is minus the covariance of its own volatility with the volatility of the log density, plus an Itô convexity term. Where HJM's derived object is an integral of the volatility across maturities, ours is η, obtained from b by the single forward recursion (7). The second is the cost. Once η is known no matrix is inverted: no Jacobian of the pricing map is ever formed and no rank condition on it appears. And since ψ is affine in β and α is bilinear in (β, ψ), the drift is a quadratic polynomial in β, as in HJM; Theorem 5.4 is what rules out the corresponding finite-time explosion.

The driftless point is the classical model: if $\beta \equiv 0$ and $\psi^1 = 0$ then $\psi \equiv 0$ and $\alpha \equiv 0$, so the discrete local variance surface is frozen, and with $\psi^1 \neq 0$ the densities still move, driven by the front alone, with $\alpha \equiv 0$ persisting.

Proposition 3.3 (Spot consistency)**.** *The spot is $S_t = K^\top q_t^j$, the same for every j by Lemma 2.3. Since $\mathbf{1}^\top A_j = \mathbf{1}^\top$, $K^\top A_j = K^\top$ and $\mathbf{1}^\top \widehat{G}_j = K^\top \widehat{G}_j = 0$, equation (7) gives*

$$\mathbf{1}^\top \psi^j = \mathbf{1}^\top \psi^1 = 0 \qquad \textit{and} \qquad K^\top \psi^j = K^\top \psi^1 = \sigma_t^S S_t \quad \textit{for every } j.$$

So the spot's volatility is the single row $K^\top\psi^1 \in \mathbb{R}^{1\times d}$, it is automatically the same at every expiry, and no further condition links the spot to the surface. Taking $\psi^1 = 0$ freezes the spot and recovers $K^\top\psi^j = 0$.

Proposition 3.3 is the discrete form of Carmona and Nadtochiy's spot consistency condition: the spot and its volatility are both *read off* the surface rather than modelled, $S = K^\top q^j$ and $\sigma^S S = K^\top\psi^j$ for any j. If $K^0 = 0$, the read-off is the zero-strike call, $S_t = C_t(0, T_j)$ for every j, as in [19] and in the multi-strike model of [15].

The drift restriction is used below through the following chain, which is what a model actually evaluates:

$$(\psi^1, b) \xrightarrow{(7)} \psi^2, \dots, \psi^M \xrightarrow{\eta=\psi/q} \alpha, \tag{11}$$

a forward march of d tridiagonal solves per expiry, $O(NMd)$ in total. The volatility structure is the only input; the drift is computed, never fitted.

4 The spot and the front expiry

The spot is not a state variable. By Proposition 3.3 it is the read-off $S_t = K^\top q_t^j$, the same scalar at every expiry. What has to be specified is the front density q^1, the initial condition of the recursion of Proposition 2.2; everything at $j \geq 2$ follows from it and from the volatility structure.

4.1 The front chart

The front density must be strictly positive and sum to one. Both are automatic in the chart

$$q^{1,i} = \frac{e^{z^i}}{\sum_k e^{z^k}}, \qquad \mathrm{d}z = m\,\mathrm{d}t + \sum_{f=1}^{d} \nu^f\,\mathrm{d}W^f, \tag{12}$$

with $z \in \mathbb{R}^{N+2}$ unconstrained. The chart is invariant under $z \mapsto z + c\mathbf{1}$, so a shift of ν along $\mathbf{1}$ leaves q^1 unchanged and only the q^1-centred part of ν is identified; write $\mathrm{dev}^f := \nu^f - (q^1)^\top\nu^f\,\mathbf{1}$ for it. The input ν is free, and plays the role b plays at $j \geq 2$. Itô's formula for a function of a vector process shows that each $q^{1,i}$ is a martingale exactly when

$$m^i = -\tfrac{1}{2}\sum_f \big(\mathrm{dev}^{i,f}\big)^2 \tag{13}$$

up to a multiple of $\mathbf{1}$. Furthermore we obtain

$$\psi^{1,\cdot,f} = q^1 \circ \mathrm{dev}^f, \qquad \eta^{1,\cdot,f} = \mathrm{dev}^f, \qquad \mathbf{1}^\top\psi^{1,\cdot,f} = 0, \tag{14}$$

with $\circ$ the entrywise product. The same computation appears in [11] and as the stochastic replicator equation [13].

Remark 4.1 (Resolution at the front expiry). Among grid measures with mean S the least dispersed is the pair of nodes straddling it, so

$$\mathrm{Var}_{q^1}(K) \;\geq\; (S - K^i)(K^{i+1} - S), \qquad K^i \leq S < K^{i+1}, \tag{15}$$

which is at most $\frac{1}{4}h^2$. The front implied volatility therefore has the floor

$$\frac{\sqrt{(S-K^i)(K^{i+1}-S)}}{S\sqrt{T_1 - t}} \;\leq\; \frac{h}{2S\sqrt{T_1 - t}},$$

largest at the midpoint of a cell and zero when the spot sits on a node, which is exactly what Proposition 4.2 forces at T_1.

4.2 Terminal settlement: pinning the spot to a strike

The chart constrains q^1 at no time $t < T_1$ beyond positivity and mass, so it imposes no terminal condition. Wissel's front parametrisation fixes the terminal state whatever the coefficients [19]; here the terminal state is a modelling choice. We take the front volatility proportional to the distance from the spot, in one factor,

$$\eta_t^{1,i} = \frac{c_t\,(K^i - S_t)}{h}, \qquad c_t = a_1\tau^{-p} \text{ near } T_1, \quad \tau = T_1 - t,\ a_1 > 0. \tag{16}$$

Only the behaviour near T_1 matters: on $[0, T_1 - \varepsilon]$ the front volatility may be any bounded, locally Lipschitz choice, and (16) need only hold on $[T_1 - \varepsilon, T_1)$. It is already centred, since $\sum_i q^{1,i}(K^i - S_t) = 0$. Its coefficients are polynomial in q^1 and bounded on $[0, T]$ for every $T < T_1$, so there is a unique strong solution on $[0, T_1)$,

$$q_t^{1,i} = q_0^{1,i}\exp\Big(\int_0^t \eta^{1,i}\,\mathrm{d}W - \tfrac{1}{2}\int_0^t (\eta^{1,i})^2\mathrm{d}s\Big), \qquad \|\eta_t^1\|_\infty \le c_t\,w/h, \quad w := K^{N+1} - K^0. \tag{17}$$

Proposition 4.2 (The front collapses onto a node, and settles exactly)**.** *If $p \ge \frac{1}{2}$ then $q_{T_1}^1 := \lim_{t\uparrow T_1} q_t^1 = e_k$ for a random index k, almost surely, so $S_{T_1} = K^k$ and*

$$\sum_i q_{T_1}^{1,i}(K^i - K^n)^+ = (S_{T_1} - K^n)^+ \qquad \textit{for every } n. \tag{18}$$

If $p < 1$ then Assumption 5.5 holds on $[0, T_1]$. The window $p \in [\frac{1}{2}, 1)$ gives both.

Proof. Each $q^{1,i}$ is a bounded martingale, so $q_t^1 \to q_{T_1}^1$ almost surely, and the variance $V_t := \sum_i q_t^{1,i}(K^i - S_t)^2$ converges to that of $q_{T_1}^1$. By (14) and (16) the volatility of $S = K^\top q^1$ is $\sum_i K^i q^{1,i}\eta^{1,i} = c_t V_t/h$ on $[T_1 - \varepsilon, T_1)$. Since S is a martingale with values in $[K^0, K^{N+1}]$,

$$\mathbb{E}\int_{T_1-\varepsilon}^{T_1} \frac{c_t^2 V_t^2}{h^2}\,\mathrm{d}t = \mathbb{E}\big[(S_{T_1} - S_{T_1-\varepsilon})^2\big] \le w^2,$$

so the integral is finite almost surely. On the event $\{V_{T_1} > 0\}$ we would have $V_t \ge V_{T_1}/2$ near T_1, and the integral would diverge, because $\int^{T_1} c_t^2\,\mathrm{d}t = a_1^2\int^{T_1}\tau^{-2p}\,\mathrm{d}t = \infty$ for $2p \ge 1$. Hence $V_{T_1} = 0$ almost surely: $q_{T_1}^1$ is a probability vector on distinct points with zero variance, that is e_k with $K^k = S_{T_1}$, and (18) reads $(K^k - K^n)^+ = (S_{T_1} - K^n)^+$. Finally η^1 is bounded on $[0, T_1 - \varepsilon]$ and $\int_{T_1-\varepsilon}^{T_1}\|\eta_t^1\|_\infty\mathrm{d}t \le (w/h)\int_{T_1-\varepsilon}^{T_1} c_t\,\mathrm{d}t < \infty$ when $p < 1$, by (17). □

The terminal spot is therefore a grid point, as in [19]; the difference is that here it is chosen, by (16), and there it is forced by the parametrisation.

Remark 4.3 (What the lattice could carry, and does not). At expiry each butterfly price $q^{j,i}$ is worth its payoff: for $S_{T_j} \in [K^k, K^{k+1}]$ the second difference defining $q^{j,i}$ telescopes to $\varpi(S_{T_j})$, the barycentric weights of the spot on the two nodes straddling it. Since $u \mapsto (u - K^n)^+$ is affine between nodes,

$$\sum_i \varpi^i(s)\,(K^i - K^n)^+ \;=\; (s - K^n)^+ \qquad \text{for every } s \in [K^0, K^{N+1}] \text{ and every model strike } K^n, \tag{19}$$

so $\varpi(s)$ is an admissible terminal state that settles every gridded contract exactly, also for s strictly inside a cell. The chart does not force the pinning; the choice (16) does, because its terminal variance must vanish. A front volatility that drives q^1 to $\varpi(s)$ with s inside a cell is left as future work (Section 6).

5 The wing constant, and existence

The log discrete local variance $\lambda = (\lambda^2, \dots, \lambda^M)$, $\lambda^j = \log \Sigma^j$, ranges over all of $\mathbb{R}^{N(M-1)}$, so positivity is automatic; the front density has its own chart (12). By Theorem 3.2 the model is the stochastic differential equation

$$\mathrm{d}\lambda_t = \Big[-\langle \eta_t, b_t\rangle - \tfrac{1}{2}|b_t|^2\Big]\mathrm{d}t + b_t\,\mathrm{d}W_t, \tag{20}$$

whose only non-elementary coefficient is $\eta = \psi/q$ from (7) and (8). Its denominator is small in the wings, and the question is whether η can blow up there.

The argument has the structure of Wissel's proof of his Theorem 3.5 [19]: log coordinates, induction over expiries, and local existence followed by a bound that rules out explosion. Its one non-standard ingredient is an a priori bound, independent of the state, on the relative volatility of the calendar spread at the second expiry. Wissel could not obtain this bound in general and imposes his condition (3.24) instead (his Remark 1 following the proof, p. 18); in our coordinates the quantity is $\eta + b$. We obtain it from an exact identity for the volatility recursion and a known inequality for inverses of M-matrices.

5.1 The volatility recursion

Write $P_j := A_j(\Sigma^j)^{-1}$, so that $q^j = P_j q^{j-1}$, and P when j is fixed.

Corollary 5.1 (Positivity, unit column sums, absorbing ends)**.** *$P \geq 0$ entrywise, $\mathbf{1}^\top P = \mathbf{1}^\top$, so $P_{im} \in [0,1]$, and the outer columns of P are e_0 and e_{N+1}.*

Proof. By (2), A_j has non-positive off-diagonal entries, and in each column the diagonal entry exceeds the sum of the moduli of the off-diagonal ones by exactly one; so A_j is a non-singular M-matrix and $P \geq 0$ [2]. The column sums follow from $\mathbf{1}^\top A_j = \mathbf{1}^\top$ (Lemma 2.3), and the outer columns of A_j are e_0 and e_{N+1}. The first two statements are also Theorem 3.3 of [8], attributed there to [1]. □

That is all that is needed about P itself. What we want next is a recursion for η alone: one step per expiry, each adding a controlled amount, so that the bound telescopes along the ladder. Dividing (7) by q^j gives

$$\eta^{j,i} \;=\; \frac{(P\psi^{j-1})_i}{q^{j,i}} \;-\; \frac{\big(P\,G_j(q^j)\beta^j\big)_i}{q^{j,i}}, \tag{21}$$

the first term carrying η^{j-1} and the second carrying b^j.

The first is an average. Writing $\psi^{j-1} = q^{j-1} \circ \eta^{j-1}$, its weights $P_{ik}q^{j-1,k}/q^{j,i}$ are non-negative because $P \geq 0$ and sum to one because $q^j = Pq^{j-1}$, so it contributes at most $\|\eta^{j-1}\|_\infty$; they are the matrix $\mathcal{A}$ of (26).

The source is the difficulty. Since G_j is a second difference, row i carries the neighbouring densities $q^{j,i\pm1}$ as well as $q^{j,i}$, so dividing by $q^{j,i}$ leaves the ratios

$$\frac{q^{j,i-1}}{q^{j,i}} \qquad \text{and} \qquad \frac{q^{j,i+1}}{q^{j,i}}, \tag{22}$$

which are unbounded in the wings, where the density decays fastest; nothing in (7) bounds them.

They are, however, an artefact of writing the row out. Since $\mathbf{1}^\top G_j(q) = 0$ by Lemma 2.3, row i of $G_j(q)\Sigma^j$ is the net outflow of node i, and by (3) that net is

$$G_j(q^j)\,\Sigma^j \;=\; A_j(\Sigma^j)q^j - q^j \;=\; q^{j-1} - q^j, \tag{23}$$

which carries no ratios at all: the individual fluxes are wing quantities, their net is not. For $b^j \equiv c$ this settles the matter, since $\beta^j = c\,\Sigma^j$ and $Pq^{j-1} = q^j$ give

$$\frac{\big(P\,G_j(q^j)\beta^j\big)_i}{q^{j,i}} = c\,\frac{\big(q^j - Pq^j\big)_i}{q^{j,i}} = c\Big(1 - \frac{(Pq^j)_i}{q^{j,i}}\Big),$$

the two ratios of (22) having collapsed into the single propagation ratio $(Pq^j)_i/q^{j,i}$. A general b only modulates the density. Let $E \in \mathbb{R}^{N\times(N+2)}$ restrict to the interior nodes, so $EE^\top = I_N$, and note that $A_j = I + \widehat{G}_j\,\mathrm{diag}(\Sigma^j)E$. Writing $E^\top(q\circ b)$ for the vector equal to q^ib^i at the interior nodes and zero at the outer ones, (4) and the commuting of diagonal matrices give

$$G_j(q)\beta^j = \widehat{G}_j\,\mathrm{diag}(\Sigma^j)E\,E^\top(q\circ b^j) = \big(A_j(\Sigma^j) - I\big)E^\top(q\circ b^j), \tag{24}$$

again the operator's own increment, applied to $E^\top(q\circ b^j)$ in place of q. The proposition is what this amounts to.

Proposition 5.2. *With $b^j = \beta^j/\Sigma^j$ and $\circ$ the entrywise product,*

$$\psi^j \;=\; P\,\psi^{j-1} \;-\; (I-P)\,E^\top\big(q^j\circ b^j\big), \tag{25}$$

equivalently, with the entrywise non-negative

$$\mathcal{A} := \mathrm{diag}(q^j)^{-1}P\,\mathrm{diag}(q^{j-1}), \qquad \mathcal{B} := \mathrm{diag}(q^j)^{-1}P\,E^\top\mathrm{diag}(Eq^j),$$

$$\eta^j \;=\; \mathcal{A}\,\eta^{j-1} \;+\; \big(\mathcal{B} - E^\top\big)b^j, \tag{26}$$

where $\mathcal{A}\mathbf{1} = \mathbf{1}$, so the transport is an averaging operator, and the i-th row of $\mathcal{B}$ sums to $(Pq^j)_i/q^{j,i} - \mathbf{1}_{\{i\in\{0,N+1\}\}}$.

Proof. By (7), $\psi^j = P\psi^{j-1} - P\,G_j(q^j)\beta^j$. Applying P to (24) at $q = q^j$ and using $PA_j = I$ gives $P\,G_j(q^j)\beta^j = (I-P)E^\top(q^j\circ b^j)$, which is (25). Dividing it by $q^{j,i}$ gives (26): the rows of $\mathcal{A}$ sum to $(Pq^{j-1})_i/q^{j,i} = 1$, and for $\mathcal{B}$ the terms $k \in \{0, N+1\}$ missing from $(Pq^j)_i$ vanish for interior i and equal $q^{j,i}$ at $k = i$ for outer i, by Corollary 5.1. □

So η^j is an average of η^{j-1} plus a source in b^j, and the march across expiries is additive.

5.2 The wing constant

That single surviving ratio, $(Pq^j)_i/q^{j,i}$, is what has to be bounded. Since $q^j = Pq^{j-1}$ it is

$$\frac{(Pq^j)_i}{q^{j,i}} = \frac{(P^2q^{j-1})_i}{(Pq^{j-1})_i}, \tag{27}$$

so what is needed is how much larger than P the square P^2 can be entrywise; and since both sides are linear and non-negative in q^{j-1}, the case $q^{j-1} = e_m$ shows that an entrywise bound is not merely sufficient but necessary. Write

$$R^{j,i} := \mathbf{1}_{\{1\le i\le N\}} + \frac{(Pq^j)_i}{q^{j,i}} - \mathbf{1}_{\{i\in\{0,N+1\}\}}, \qquad \Theta_j := \max_i R^{j,i}, \qquad \Xi_j := \sum_{r=1}^{N} P_{rr} \le N, \tag{28}$$

and $H_j := \max_i|\eta^{j,i}|$. Here $R^{j,i}$ is the row bound of (26) at node i, and Θ_j, the *wing constant*, is its size across the slab. The theorem shows Θ_j is bounded by the grid alone, uniformly in the state; the bound rests on a single inequality for inverses of M-matrices.

Lemma 5.3 (Squared propagator)**.** *For all nodes i, r, m,*

$$P_{ir}\,P_{rm} \;\le\; P_{im}\,P_{rr}, \tag{29}$$

and consequently, for all i, m,

$$\big(P^2\big)_{im} \le \begin{cases} \Xi_j\,P_{im}, & 1\le i\le N,\\ (1+\Xi_j)\,P_{im}, & i\in\{0,N+1\}. \end{cases} \tag{30}$$

Proof. P is the inverse of an M-matrix by Corollary 5.1, and every inverse M-matrix is a path product matrix, $a_{ij}a_{jk} \le a_{ik}a_{jj}$ for all i, j, k [12, Theorem 4.5]; (29) is the case $(i, j, k) = (i, r, m)$. Summing it over r gives (30): an outer r contributes only if $r = i$, since column r of P is e_r, and then contributes P_{im}. □

Theorem 5.4. *For every $\Sigma \in (0,\infty)^N$,*

$$H_j \;\le\; H_{j-1} + \Theta_j\|b^j\|_\infty, \tag{31}$$

and

$$\Theta_j \le 1+\Xi_j \le N+1, \qquad \Theta^* := \sup_{\Sigma}\sum_{j=2}^{M}\Theta_j \le (M-1)(N+1). \tag{32}$$

Proof. Since $q^j = Pq^{j-1} \ge 0$, Lemma 5.3 gives $(Pq^j)_i = \sum_m (P^2)_{im}q^{j-1,m} \le \Xi_j\,q^{j,i}$ for interior i and $\le (1+\Xi_j)q^{j,i}$ for outer i, so $R^{j,i} \le 1+\Xi_j$ in both cases. Finally, $\mathcal{A} \ge 0$ with $\mathcal{A}\mathbf{1} = \mathbf{1}$ gives $\|\mathcal{A}\eta^{j-1}\|_\infty \le H_{j-1}$, while row i of $(\mathcal{B} - E^\top)b^j$ has modulus at most $\|b^j\|_\infty R^{j,i}$ by the row sums of Proposition 5.2. □

5.3 Existence

Iterating (31) from the front slice,

$$\sup_{j,i}|\eta_t^{j,i}| \;\le\; \|\eta_t^1\|_\infty + (M-1)(N+1)\,\|b\|_\infty, \tag{33}$$

and iterating (25) shows that η^j is affine in η^1, with coefficients depending on λ only through $P_2,\dots,P_j$ and $q^2,\dots,q^j$. The only t on the right of (33) is inside η_t^1, which enters linearly, so it has to be integrable in time rather than bounded.

Assumption 5.5 (Integrable front volatility)**.** For a horizon $T \le T_1$,

$$\int_0^T \|\eta_t^1\|_\infty \,\mathrm{d}t \;<\; \infty \qquad \text{almost surely.} \tag{34}$$

Theorem 5.6 (Existence)**.** *Let the front volatility $\nu = \nu(t,z)$ in* (12) *and the volatility $b = b(t,z,\lambda)$ be locally Lipschitz in (t,z,λ) on $[0,T_1)$, with $|b| \le B$, and ν bounded on $[0,T]$ for every $T < T_1$.*

(i) For every $T < T_1$ the front chart (12)–(13) *together with* (20) *has a unique strong solution (z,λ) on $[0,T]$, and*

$$0 \;<\; \inf_{t\le T}\min_{j,i}\Sigma_t^{j,i} \;\le\; \sup_{t\le T}\max_{j,i}\Sigma_t^{j,i} \;<\; \infty \qquad \textit{almost surely.} \tag{35}$$

(ii) If moreover Assumption 5.5 holds with $T = T_1$, then λ_t has a finite limit λ_{T_1} as $t \uparrow T_1$, and (35) *holds with $T = T_1$.*

No condition is imposed on Σ, on the wings, or on the number of factors.

Proof. (i) Fix $T < T_1$ and let B_ν bound $|\nu|$ on $[0,T]$. The coefficients are locally Lipschitz: $q^1 = \mathrm{softmax}(z)$ and $\Sigma = e^\lambda$ are positive, so each A_j is a non-singular M-matrix and everything except ν and b is smooth in (z,λ). Adding time as a state variable to cover the time dependence, [14, Theorem V.38, p. 303] gives a unique solution up to an explosion time ζ, with $\limsup_{t\to\zeta}\|(z_t,\lambda_t)\| = \infty$ on $\{\zeta < \infty\}$. The drift of λ is bounded by (33), since $\|\eta^1\|_\infty = \|\mathrm{dev}\|_\infty \le 2B_\nu$, and the drift m of z is bounded by $2B_\nu^2$; let D bound both. On $[0,\zeta\wedge T)$ the drift part of (z,λ) is therefore at most DT, and the noise part, having integrand bounded by $B_\nu + B$, is a martingale bounded in L^2, hence convergent as $t \to \zeta\wedge T$. So (z,λ) stays bounded on $[0,\zeta\wedge T)$, which rules out $\zeta \le T$. The solution exists on $[0,T]$, λ is bounded there, and (35) follows from $\Sigma = e^\lambda$.

(ii) By (i) the solution exists on $[0,T_1)$. By (33) the drift of λ is bounded by $g(t) := \big(\|\eta_t^1\|_\infty + (M-1)(N+1)B\big)B + \frac{1}{2}B^2$, which is integrable on $[0,T_1)$ by Assumption 5.5, so the drift part of λ converges as $t \uparrow T_1$. The noise part $\int_0^t b\,\mathrm{d}W$ has integrand bounded by B, so it is a martingale bounded in L^2 on $[0,T_1)$ and converges as well. Hence λ_t has a finite limit, λ is bounded on $[0,T_1]$, and (35) holds with $T = T_1$. Nothing is claimed for z, which may run off at T_1 as the front collapses. □

The settling front (16) is bounded on every $[0,T]$ with $T < T_1$ by (17), and satisfies Assumption 5.5 with $T = T_1$ for $p < 1$ (Proposition 4.2). So for $p \in [\frac{1}{2}, 1)$ the code-book reaches its first expiry with Σ bounded away from 0 and ∞ and every gridded contract settling at its payoff.

5.4 Beyond the first expiry

At T_1 the front expires and the next expiry becomes the front. The settling of Proposition 4.2 makes this switch clean: the spot sits on a node, and the new front density is a column of the propagator. Indeed, if $q^l_{T_l} = e_k$ then the code-book $A_{l+1}(\Sigma^{l+1})\, q^{l+1} = q^l$ gives

$$q^{l+1}_{T_l} = P_{l+1}\, e_k, \qquad P_{l+1} := A_{l+1}\big(\Sigma^{l+1}_{T_l}\big)^{-1}, \tag{36}$$

the spread of a unit mass at the settled spot by one step of the code-book. To restart the front chart we need this density to be strictly positive.

Lemma 5.7 (The new front)**.** *Let $P = A_j(\Sigma^j)^{-1}$ with $\Sigma^j \in (0,\infty)^N$. If k is an interior node, then $(Pe_k)_i > 0$ for every node i and $K^\top P e_k = K^k$. If $k \in \{0, N+1\}$, then $Pe_k = e_k$.*

Proof. Order the nodes with the interior first. The outer columns of A_j are e_0 and e_{N+1} (Corollary 5.1), so

$$A_j = \begin{pmatrix} A_{II} & 0 \\ A_{BI} & I \end{pmatrix}, \qquad P = \begin{pmatrix} A_{II}^{-1} & 0 \\ -A_{BI}A_{II}^{-1} & I \end{pmatrix}.$$

By (2), A_{II} is tridiagonal with strictly negative off-diagonal entries, hence an irreducible non-singular M-matrix, so $A_{II}^{-1} > 0$ entrywise [2]. The block $-A_{BI} \geq 0$ has a positive entry in the row of K^0 (column K^1) and in the row of K^{N+1} (column K^N), so the outer entries of an interior column of P are positive as well. The identity $K^\top P = K^\top$ follows from $K^\top A_j = K^\top$ (Lemma 2.3), and the outer columns are Corollary 5.1. □

Corollary 5.8 (Existence on the whole ladder)**.** *Let b be bounded and locally Lipschitz. On each interval $[T_{l-1}, T_l]$, $l = 1, \dots, M$, let the current front density q^l follow the chart* (12) *with the settling volatility* (16)*, where $\tau = T_l - t$ and $p \in [\frac{1}{2}, 1)$. Then the model has a solution on $[0, T_M]$, unique given these choices, such that*

(i) every gridded price is a continuous martingale up to its expiry;

(ii) at every T_l the spot sits on a node K^{k_l} and every contract expiring at T_l settles at its payoff;

(iii) if k_l is interior, the model continues on $[T_l, T_{l+1}]$ from the finite state $z = \log q^{l+1}_{T_l}$ and $\lambda^{l+2}, \dots, \lambda^M$ at T_l; if k_l is an outer node, the spot is absorbed there: $q^j_t = e_{k_l}$ for all $j > l$ and $t \geq T_l$.

Proof. By induction on l. On $[0, T_1]$ this is Proposition 4.2 with Theorem 5.6, whose part (ii) also gives the limit λ_{T_1}, finite. Suppose the solution exists on $[0, T_l]$ with λ_{T_l} finite and $q^l_{T_l} = e_k$.

If k is interior, (36) and Lemma 5.7 give a strictly positive new front, so $z = \log q^{l+1}_{T_l}$ is finite, and $K^\top q^{l+1}_{T_l} = K^k = S_{T_l}$, so the spot is continuous. On $[T_l, T_{l+1}]$ the ladder is that of Sections 3–5 with front q^{l+1} and expiries $T_{l+1} < \cdots < T_M$, started from the $\mathcal{F}_{T_l}$-measurable state $(z, \lambda^{l+2}, \dots, \lambda^M)$. Theorem 5.6 applies with this $\mathcal{F}_{T_l}$-measurable initial condition, the solution of [14, Theorem V.38] depending continuously on its starting point, and with Proposition 4.2 and part (ii) of that theorem gives a unique strong solution on $[T_l, T_{l+1}]$ that settles at T_{l+1} with $\lambda_{T_{l+1}}$ finite. The densities q^j, $j > l+1$, are given by the same recursion on both sides of T_l with continuous inputs, so they are continuous at T_l.

If k is an outer node, Lemma 5.7 gives $q^j_{T_l} = e_k$ for every $j > l$. The constant process $q^j_t \equiv e_k$ satisfies every recursion, since $A_j e_k = e_k$, and is a martingale, so all prices are constant and settle at their payoffs; we freeze λ at λ_{T_l}, on which no price then depends.

Uniqueness on each interval gives uniqueness on $[0, T_M]$. □

6 What is open

1. *An off-lattice terminal spot.* By Remark 4.3 the chart represents every $s \in [K^0, K^{N+1}]$ at expiry, and the state that does so is $\varpi(s)$, carried by the two nodes straddling s. Whether a front volatility exists driving q^1 to $\varpi(s)$ for s strictly inside a cell, rather than to a single node as (16) does, is open. The obstruction is that exact settlement forces $q^1_{T_1} = \varpi(S_{T_1})$ and hence $\int_0^{T_1} (\eta^{1,i})^2 \mathrm{d}t < \infty$ for both surviving nodes, which by the argument of Proposition 4.2 they cannot both satisfy when the volatility is proportional to distance from the spot. Until it is answered the model of Proposition 4.2 pins, as [19] does, and his question of a front that does not pin the spot remains open.

2. *The continuum limit.* The constant $(M-1)(N+1)$ grows with the grid. Whether $\sup_N \Theta_j < \infty$ on a class of realistic surfaces, which is what a continuum statement would need, is open.

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